

# **ONLINE RETAIL DATA ANALYSIS USING HADOOP**

Big Data Analytics Lab

**Submitted by:**

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## Problem Statement

Analyse the UCI Online Retail transaction dataset using Hadoop MapReduce and Apache Pig, to determine which products, customers and countries generate the most revenue and how sales behave over time.

## Dataset Description

Dataset: Online Retail, UCI Machine Learning Repository

Source: <https://archive.ics.uci.edu/dataset/352/online+retail>

The dataset records every transaction of a UK-based online gift retailer between 1 December 2010 and 9 December 2011, a total of 541,909 rows. The publisher distributes it as a single Excel workbook; because Hadoop reads plain text, the workbook was converted to CSV and partitioned by invoice month into 13 files, so that HDFS produces 13 input splits and the jobs run 13 map tasks in parallel.

|             |                                                     |
|-------------|-----------------------------------------------------|
| InvoiceNo   | invoice number; a leading C marks a cancelled order |
| StockCode   | product identifier                                  |
| Description | product name                                        |
| Quantity    | units sold; negative on returns                     |
| InvoiceDate | date and time of the transaction                    |
| UnitPrice   | price per unit in pounds sterling                   |
| CustomerID  | customer identifier; empty on 24.9% of rows         |
| Country     | country of the customer                             |

Preparation: commas inside Description were replaced with spaces (4,796 rows) and quote characters removed (1,239 rows), so that every field splits cleanly on a comma. Country contains no commas. After conversion all 541,909 rows split into exactly eight fields.

## **Summary of Data Analysis**

**The following MapReduce design patterns are implemented:**

1. Filtering
2. Summarization
3. Data Organization
4. Top-N selection

**The following MapReduce analyses are performed:**

1. Total sales revenue by product
2. Average invoice value, with per-invoice detail
3. Top 10 customers by total spending
4. Binning of transactions by country

**The following Pig analyses are performed:**

1. Top 5 countries by revenue
2. Top 10 products by revenue
3. Top 10 products in each country
4. Top 10 customers by spending
5. Monthly sales trend

## Map Reduce Analysis

### Total Sales by Product

Returns the total revenue generated by each product, calculated as Quantity multiplied by UnitPrice and summed per StockCode. Cancelled invoices are excluded.

1. Driver Class : sets up configuration for the Mapper, Combiner and Reducer
2. Mapper Class : emits StockCode of type Text and line revenue of type DoubleWritable
3. Reducer Class : emits StockCode as key and its total revenue of type DoubleWritable

### Driver Class

```
1  package totalsalesbyproduct;
2
3  import java.io.IOException;
4
5  import org.apache.hadoop.conf.Configuration;
6  import org.apache.hadoop.fs.Path;
7  import org.apache.hadoop.io.DoubleWritable;
8  import org.apache.hadoop.io.Text;
9  import org.apache.hadoop.mapreduce.Job;
10 import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
11 import org.apache.hadoop.mapreduce.lib.input.TextInputFormat;
12 import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;
13 import org.apache.hadoop.mapreduce.lib.output.TextOutputFormat;
14
15 /**
16  * Total Sales by Product : returns the total revenue generated by every product,
17  * ignoring cancelled invoices. Sets up configuration for the Mapper, Combiner
18  * and Reducer classes.
19  */
20 public class DriverClass {
21
22     public static void main(String[] args)
23         throws IOException, InterruptedException, ClassNotFoundException {
24
```

```

25     Configuration conf = new Configuration();
26     Job job = Job.getInstance(conf, "Total Sales By Product");
27     job.setJarByClass(DriverClass.class);
28
29     job.setMapperClass(MapperClass.class);
30     job.setCombinerClass(ReducerClass.class);
31     job.setReducerClass(ReducerClass.class);
32
33     job.setMapOutputKeyClass(Text.class);
34     job.setMapOutputValueClass(DoubleWritable.class);
35     job.setOutputKeyClass(Text.class);
36     job.setOutputValueClass(DoubleWritable.class);
37
38     job.setInputFormatClass(TextInputFormat.class);
39     job.setOutputFormatClass(TextOutputFormat.class);
40
41     FileInputFormat.addInputPath(job, new Path(args[0]));
42     FileOutputFormat.setOutputPath(job, new Path(args[1]));
43
44     System.exit(job.waitForCompletion(true) ? 0 : 1);
45 }
46 }

```

## Mapper Class

```

1  package totalsalesbyproduct;
2
3  import java.io.IOException;
4
5  import org.apache.hadoop.io.DoubleWritable;
6  import org.apache.hadoop.io.LongWritable;
7  import org.apache.hadoop.io.Text;
8  import org.apache.hadoop.mapreduce.Mapper;
9
10 /**
11  * Emits StockCode of type Text and the line revenue (Quantity * UnitPrice) of
12  * type DoubleWritable.
13  *
14  * Dataset columns: 0 InvoiceNo, 1 StockCode, 2 Description, 3 Quantity,
15  * 4 InvoiceDate, 5 UnitPrice, 6 CustomerID, 7 Country.
16  */
17 public class MapperClass extends Mapper<LongWritable, Text, Text, DoubleWritable> {
18
19     private Text stockCode = new Text();
20     private DoubleWritable revenue = new DoubleWritable();
21
22     @Override
23     public void map(LongWritable key, Text value, Context context)
24         throws IOException, InterruptedException {

```

```

25
26     String[] col = value.toString().split(",");
27     if (col.length < 8) {
28         return;
29     }
30
31     // An InvoiceNo beginning with C marks a cancelled order; those rows carry
32     // negative quantities and would subtract from genuine sales.
33     if (col[0].startsWith("C") || col[0].startsWith("c")) {
34         context.getCounter("Retail", "CancelledRowsSkipped").increment(1);
35         return;
36     }
37
38     try {
39         double qty = Double.parseDouble(col[3]);
40         double price = Double.parseDouble(col[5]);
41         if (qty <= 0 || price <= 0) {
42             context.getCounter("Retail", "NonPositiveRowsSkipped").increment(1);
43             return;
44         }
45         stockCode.set(col[1]);
46         revenue.set(qty * price);
47         context.write(stockCode, revenue);
48     } catch (NumberFormatException e) {
49         context.getCounter("Retail", "UnparseableRowsSkipped").increment(1);
50     }
51 }
52 }

```

## Reducer Class

```

1  package totalsalesbyproduct;
2
3  import java.io.IOException;
4
5  import org.apache.hadoop.io.DoubleWritable;
6  import org.apache.hadoop.io.Text;
7  import org.apache.hadoop.mapreduce.Reducer;
8
9  /**
10   * Emits StockCode as key of type Text and its total revenue of type
11   * DoubleWritable. Also usable as the Combiner, since summing is associative.
12   */
13  public class ReducerClass extends Reducer<Text, DoubleWritable, Text, DoubleWritable> {
14
15      private DoubleWritable total = new DoubleWritable();
16
17      @Override
18      public void reduce(Text key, Iterable<DoubleWritable> values, Context context)
19          throws IOException, InterruptedException {
20
21          double sum = 0;
22          for (DoubleWritable value : values) {
23              sum += value.get();
24          }
25          total.set(Math.round(sum * 100.0) / 100.0);
26          context.write(key, total);
27      }
28  }

```

## Execution

```
YTRUN-02_total_sales_by_product

D:\Desktop\bdal_project>javac -classpath C:\hadoop\share\hadoop\common\*;C:\hadoop\share\hadoop\mapreduce\
*;C:\hadoop\share\hadoop\hdfs\*;C:\hadoop\share\hadoop\common\lib* -d D:\Desktop\bdal_project\TotalSalesB
yProduct\target\classes D:\Desktop\bdal_project\TotalSalesByProduct\src\main\java\totalsalesbyproduct\Driv
erClass.java D:\Desktop\bdal_project\TotalSalesByProduct\src\main\java\totalsalesbyproduct\MapperClass.jav
a D:\Desktop\bdal_project\TotalSalesByProduct\src\main\java\totalsalesbyproduct\ReducerClass.java

D:\Desktop\bdal_project>jar -cvf D:\Desktop\bdal_project\TotalSalesByProduct\target\TotalSalesByProduct.jar
D:\Desktop\bdal_project\TotalSalesByProduct\target\classes .
added manifest
adding: totalsalesbyproduct/(in = 0) (out= 0)(stored 0%)
adding: totalsalesbyproduct/DriverClass.class(in = 1843) (out= 918)(deflated 56%)
adding: totalsalesbyproduct/MapperClass.class(in = 2471) (out= 1081)(deflated 56%)
adding: totalsalesbyproduct/ReducerClass.class(in = 1798) (out= 776)(deflated 56%)

D:\Desktop\bdal_project>hadoop jar D:\Desktop\bdal_project\TotalSalesByProduct\target\TotalSalesByProduct.jar
totalsalesbyproduct.DriverClass /BdalProject/data /BdalProject/output/totalsales
2026-09-08 17:42:33,567 INFO client.RMProxy: Connecting to ResourceManager at /0.0.0.0:8032
2026-09-08 17:42:34,722 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not perform
ed. Implement the Tool interface and execute your application with ToolRunner to remedy this.
2026-09-08 17:42:34,738 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop
-yarn/staging/HP/.staging/job_1788866353307_0007
2026-09-08 17:42:35,027 INFO Input.FileInputFormat: Total input files to process : 13
2026-09-08 17:42:35,125 INFO mapreduce.JobSubmitter: number of splits:13
2026-09-08 17:42:35,287 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1788866353307_0007
2026-09-08 17:42:35,291 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-09-08 17:42:35,509 INFO conf.Configuration: resource-types.xml not found
2026-09-08 17:42:35,510 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2026-09-08 17:42:35,603 INFO impl.YarnClientImpl: Submitted application application_1788866353307_0007
2026-09-08 17:42:35,667 INFO mapreduce.Job: The url to track the job: http://DESKTOP-S64ND9J:8088/proxy/ap
plication_1788866353307_0007/
2026-09-08 17:42:35,669 INFO mapreduce.Job: Running job: job_1788866353307_0007
2026-09-08 17:42:53,028 INFO mapreduce.Job: Job job_1788866353307_0007 running in uber mode : false
2026-09-08 17:42:53,029 INFO mapreduce.Job: map 0% reduce 0%
2026-09-08 17:43:15,783 INFO mapreduce.Job: map 8% reduce 0%
2026-09-08 17:43:18,993 INFO mapreduce.Job: map 23% reduce 0%
2026-09-08 17:43:21,117 INFO mapreduce.Job: map 31% reduce 0%
2026-09-08 17:43:26,417 INFO mapreduce.Job: map 38% reduce 0%
2026-09-08 17:43:27,466 INFO mapreduce.Job: map 46% reduce 0%
2026-09-08 17:43:30,522 INFO mapreduce.Job: map 54% reduce 0%
2026-09-08 17:43:47,009 INFO mapreduce.Job: map 62% reduce 0%
2026-09-08 17:43:48,035 INFO mapreduce.Job: map 69% reduce 0%
2026-09-08 17:43:50,107 INFO mapreduce.Job: map 77% reduce 0%
```

```
YTRUN-02_total_sales_by_product

10125 994.84
10133 1544.22
10135 2207.39
11001 2389.44
15030 41.47
15034 840.02
15036 18569.84
15039 1979.49
15044A 1456.72
15044B 946.83
15044C 1025.94
15044D 1822.71
15056BL 15828.84
15056N 22176.49
15056P 4366.82
15056bl 1086.88
15056n 1585.28
15056p 412.48
15058A 1668.49
15058B 1885.27
15058C 1429.82
15060B 1807.55
15060b 24.87
16008 360.71
16010 3.6
16011 372.33
16012 261.24
16014 4335.76
16015 289.98
16016 1053.45
16020C 69.9
16033 100.8
16043 11.76
16045 380.0
16046 57.8
16048 205.42
16049 82.08
16052 115.3
16054 140.52
16151A 61.25
16156L 199.5
16156S 2331.0
16161G 70.0
```

| YTRUN-02_total_sales_by_product |         |
|---------------------------------|---------|
| 16161G                          | 70.0    |
| 16161M                          | 10.5    |
| 16161P                          | 2990.92 |
| 16161U                          | 1687.5  |
| 16162L                          | 13.0    |
| 16162M                          | 11.76   |
| 16168M                          | 173.12  |
| 16169E                          | 1892.46 |
| 16169K                          | 84.0    |
| 16169M                          | 277.2   |
| 16169N                          | 10.5    |
| 16169P                          | 21.0    |
| 16202A                          | 3.5     |
| 16202B                          | 14.5    |
| 16202E                          | 7.5     |
| 16206B                          | 8.75    |
| 16207A                          | 163.1   |
| 16207B                          | 76.14   |
| 16216                           | 234.0   |
| 16218                           | 283.31  |
| 16219                           | 407.72  |
| 16225                           | 691.85  |
| 16235                           | 702.96  |
| 16236                           | 320.09  |
| 16237                           | 1048.56 |
| 16238                           | 322.09  |
| 16244B                          | 7.85    |
| 16248B                          | 16.25   |
| 16254                           | 42.58   |
| 16258A                          | 1       |

## Average Invoice Value

Returns the number of order lines, the total value and the mean line value of every invoice, together with the average invoice value across the whole dataset.

1. Driver Class : sets up configuration for the Mapper, Combiner, Reducer and Tuple class
2. Mapper Class : emits InvoiceNo of type Text and a Tuple of custom type InvoiceValueTuple
3. Combiner Class : merges the order lines of one invoice on the map side
4. Reducer Class : emits InvoiceNo as key and the summarised Tuple, and writes the overall average from cleanup()
5. Tuple Class : POJO carrying the line count and their summed value

## Driver Class

```
1 package averageinvoicevalue;
2
3 import java.io.IOException;
4
5 import org.apache.hadoop.conf.Configuration;
6 import org.apache.hadoop.fs.Path;
7 import org.apache.hadoop.io.Text;
8 import org.apache.hadoop.mapreduce.Job;
9 import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
10 import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;
11
12 /**
13  * Average Invoice Value : returns the line count, total value and mean line
14  * value of every invoice, and the average invoice value across the whole
15  * dataset. Sets up configuration for the Mapper, Combiner, Reducer and Tuple
16  * class.
17  */
18 public class DriverClass {
19
20     public static void main(String[] args)
21         throws IOException, InterruptedException, ClassNotFoundException {
22
23         Configuration conf = new Configuration();
24         Job job = Job.getInstance(conf, "Average Invoice Value");
```

```
25         job.setJarByClass(DriverClass.class);
26
27         job.setMapperClass(MapperClass.class);
28         job.setCombinerClass(CombinerClass.class);
29         job.setReducerClass(ReducerClass.class);
30
31         job.setMapOutputKeyClass(Text.class);
32         job.setMapOutputValueClass(InvoiceValueTuple.class);
33         job.setOutputKeyClass(Text.class);
34         job.setOutputValueClass(InvoiceValueTuple.class);
35
36         // A single reducer, so the dataset-wide average written from cleanup()
37         // is computed once over every invoice rather than once per partition.
38         job.setNumReduceTasks(1);
39
40         FileInputFormat.addInputPath(job, new Path(args[0]));
41         FileOutputFormat.setOutputPath(job, new Path(args[1]));
42
43         System.exit(job.waitForCompletion(true) ? 0 : 1);
44     }
45 }
```

## Mapper Class

```
1  package averageinvoicevalue;
2
3  import java.io.IOException;
4
5  import org.apache.hadoop.io.LongWritable;
6  import org.apache.hadoop.io.Text;
7  import org.apache.hadoop.mapreduce.Mapper;
8
9  /**
10   * Emits InvoiceNo of type Text and a Tuple of custom type InvoiceValueTuple
11   * holding one order line and its value.
12   */
13  public class MapperClass extends Mapper<LongWritable, Text, Text, InvoiceValueTuple> {
14
15      private Text invoiceNo = new Text();
16      private InvoiceValueTuple outTuple = new InvoiceValueTuple();
17
18      @Override
19      public void map(LongWritable key, Text value, Context context)
20          throws IOException, InterruptedException {
21
22          String[] col = value.toString().split(",");
23          if (col.length < 8) {
24              return;
```

```
25      }
26      if (col[0].startsWith("C") || col[0].startsWith("c")) {
27          context.getCounter("Retail", "CancelledRowsSkipped").increment(1);
28          return;
29      }
30
31      try {
32          double qty = Double.parseDouble(col[3]);
33          double price = Double.parseDouble(col[5]);
34          if (qty <= 0 || price <= 0) {
35              return;
36          }
37          invoiceNo.set(col[0]);
38          outTuple.setLineCount(1);
39          outTuple.setTotalAmount(qty * price);
40          context.write(invoiceNo, outTuple);
41      } catch (NumberFormatException e) {
42          context.getCounter("Retail", "UnparseableRowsSkipped").increment(1);
43      }
44  }
45  }
```

## Combiner Class

```
1 package averageinvoicevalue;
2
3 import java.io.IOException;
4
5 import org.apache.hadoop.io.Text;
6 import org.apache.hadoop.mapreduce.Reducer;
7
8 /**
9  * Combiner Class : merges the order lines of one invoice on the map side.
10  * Line count and amount are both carried forward so the reducer still sees a
11  * faithful partial sum.
12  */
13 public class CombinerClass extends Reducer<Text, InvoiceValueTuple, Text, InvoiceValueTuple> {
14
15     private InvoiceValueTuple result = new InvoiceValueTuple();
16
17     @Override
18     protected void reduce(Text key, Iterable<InvoiceValueTuple> values, Context context)
19         throws IOException, InterruptedException {
20
21         int lines = 0;
22         double amount = 0;
23         for (InvoiceValueTuple val : values) {
24             lines += val.getLineCount();
25             amount += val.getTotalAmount();
26         }
27         result.setLineCount(lines);
28         result.setTotalAmount(amount);
29         context.write(key, result);
30     }
31 }
```

## Reducer Class

```
1 package averageinvoicevalue;
2
3 import java.io.IOException;
4
5 import org.apache.hadoop.io.Text;
6 import org.apache.hadoop.mapreduce.Reducer;
7
8 /**
9  * Emits InvoiceNo as key of type Text and a Tuple of custom type
10  * InvoiceValueTuple giving the line count, invoice total and mean line value.
11  *
12  * The overall average invoice value cannot be produced per key, so the running
13  * totals are held across reduce() calls and written once from cleanup().
14  */
15 public class ReducerClass extends Reducer<Text, InvoiceValueTuple, Text, InvoiceValueTuple> {
16
17     private InvoiceValueTuple result = new InvoiceValueTuple();
18     private int invoiceCount = 0;
19     private double grandTotal = 0;
20
21     @Override
22     protected void reduce(Text key, Iterable<InvoiceValueTuple> values, Context context)
23         throws IOException, InterruptedException {
24
```

```

25     int lines = 0;
26     double amount = 0;
27     for (InvoiceValueTuple val : values) {
28         lines += val.getLineCount();
29         amount += val.getTotalAmount();
30     }
31     result.setLineCount(lines);
32     result.setTotalAmount(amount);
33     context.write(key, result);
34
35     invoiceCount++;
36     grandTotal += amount;
37 }
38
39 @Override
40 protected void cleanup(Context context) throws IOException, InterruptedException {
41     if (invoiceCount == 0) {
42         return;
43     }
44     InvoiceValueTuple overall = new InvoiceValueTuple();
45     overall.setLineCount(invoiceCount);
46     overall.setTotalAmount(grandTotal);
47     context.write(new Text("ZZZ_AVERAGE_INVOICE_VALUE"), overall);
48 }
49 }

```

## Tuple Class

```

1  package averageinvoicevalue;
2
3  import java.io.DataInput;
4  import java.io.DataOutput;
5  import java.io.IOException;
6
7  import org.apache.hadoop.io.Writable;
8
9  /**
10   * Tuple Class : POJO carrying the number of order lines and their summed value.
11   * Both parts must travel together, otherwise the Combiner cannot merge partial
12   * results without distorting the average.
13   */
14  public class InvoiceValueTuple implements Writable {
15
16     private int lineCount = 0;
17     private double totalAmount = 0;
18
19     public int getLineCount() {
20         return lineCount;
21     }
22
23     public void setLineCount(int lineCount) {
24         this.lineCount = lineCount;

```

```

25     }
26
27     public double getTotalAmount() {
28         return totalAmount;
29     }
30
31     public void setTotalAmount(double totalAmount) {
32         this.totalAmount = totalAmount;
33     }
34
35     public void write(DataOutput d) throws IOException {
36         d.writeInt(lineCount);
37         d.writeDouble(totalAmount);
38     }
39
40     public void readFields(DataInput di) throws IOException {
41         lineCount = di.readInt();
42         totalAmount = di.readDouble();
43     }
44
45     @Override
46     public String toString() {
47         double avg = (lineCount == 0) ? 0 : totalAmount / lineCount;
48         return lineCount + "\t"
49             + (Math.round(totalAmount * 100.0) / 100.0) + "\t"
50             + (Math.round(avg * 100.0) / 100.0);
51     }
52 }

```

## Execution

```

YTRUN-03_average_invoice_value

D:\Desktop\bdal_project>javac -classpath C:\hadoop\share\hadoop\common\*;C:\hadoop\share\hadoop\mapreduce\
*;C:\hadoop\share\hadoop\hdfs\*;C:\hadoop\share\hadoop\common\lib\* -d D:\Desktop\bdal_project\AverageInvo
iceValue\target\classes D:\Desktop\bdal_project\AverageInvoiceValue\src\main\java\averageinvoicevalue\Driv
erClass.java D:\Desktop\bdal_project\AverageInvoiceValue\src\main\java\averageinvoicevalue\MapperClass.jav
a D:\Desktop\bdal_project\AverageInvoiceValue\src\main\java\averageinvoicevalue\CombinerClass.java D:\Desk
top\bdal_project\AverageInvoiceValue\src\main\java\averageinvoicevalue\ReducerClass.java D:\Desktop\bdal_p
roject\AverageInvoiceValue\src\main\java\averageinvoicevalue\InvoiceValueTuple.java

D:\Desktop\bdal_project>jar -cvf D:\Desktop\bdal_project\AverageInvoiceValue\target\AverageInvoiceValue.jar
r -C D:\Desktop\bdal_project\AverageInvoiceValue\target\classes .
added manifest
adding: averageinvoicevalue/(in = 0) (out= 0)(stored 0%)
adding: averageinvoicevalue/CombinerClass.class(in = 1869) (out= 801)(deflated 57%)
adding: averageinvoicevalue/DriverClass.class(in = 1717) (out= 892)(deflated 48%)
adding: averageinvoicevalue/InvoiceValueTuple.class(in = 1478) (out= 804)(deflated 45%)
adding: averageinvoicevalue/MapperClass.class(in = 2495) (out= 1100)(deflated 55%)
adding: averageinvoicevalue/ReducerClass.class(in = 2435) (out= 1018)(deflated 58%)

D:\Desktop\bdal_project>hadoop jar D:\Desktop\bdal_project\AverageInvoiceValue\target\AverageInvoiceValue.
jar averageinvoicevalue.DriverClass /BdalProject/data /BdalProject/output/avginvoice
2026-09-08 17:46:09,364 INFO client.RMProxy: Connecting to Resource Manager at /0.0.0.0:8032
2026-09-08 17:46:10,859 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not perform
ed. Implement the Tool interface and execute your application with ToolRunner to remedy this.
2026-09-08 17:46:10,875 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop
-yarn/staging/HP/.staging/job_1788866353307_0008
2026-09-08 17:46:11,412 INFO Input.FileInputFormat: Total input files to process : 13
2026-09-08 17:46:11,715 INFO mapreduce.JobSubmitter: number of splits:13
2026-09-08 17:46:12,350 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1788866353307_0008
2026-09-08 17:46:12,373 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-09-08 17:46:13,163 INFO conf.Configuration: resource-types.xml not found
2026-09-08 17:46:13,165 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2026-09-08 17:46:13,276 INFO impl.YarnClientImpl: Submitted application application_1788866353307_0008
2026-09-08 17:46:13,346 INFO mapreduce.Job: The url to track the job: http://DESKTOP-564ND9J:8088/proxy/ap
plication_1788866353307_0008/
2026-09-08 17:46:13,347 INFO mapreduce.Job: Running job: job_1788866353307_0008
2026-09-08 17:46:34,824 INFO mapreduce.Job: Job job_1788866353307_0008 running in uber mode : false
2026-09-08 17:46:34,826 INFO mapreduce.Job: map 0% reduce 0%
2026-09-08 17:46:57,468 INFO mapreduce.Job: map 8% reduce 0%
2026-09-08 17:47:01,601 INFO mapreduce.Job: map 15% reduce 0%
2026-09-08 17:47:04,747 INFO mapreduce.Job: map 23% reduce 0%
2026-09-08 17:47:05,948 INFO mapreduce.Job: map 31% reduce 0%
2026-09-08 17:47:07,051 INFO mapreduce.Job: map 38% reduce 0%

```

```
YTRUN-03_average_invoice_value
536416 6      225.7  37.62
536420 14     233.45 16.68
536423 9      303.9  33.77
536425
D:\Desktop\bdal_project>hdfs dfs -tail /BdalProject/output/avginvoice/part-r-00000
2618.89 40.29
581477 10     291.78 29.18
581478 18     391.02 21.72
581479 1      29.5   29.5
581480 9      305.7  33.97
581481 16     242.3  15.14
581482 6      488.4  81.4
581483 1      168469.6 168469.6
581485 3      741.9  247.3
581486 16     407.68 25.48
581487 1      406.8  406.8
581488 3      415.6  138.53
581489 2      171.6  85.8
581491 1      19.8   19.8
581492 731    7689.23 10.52
581493 17     208.64 12.27
581494 18     339.14 18.84
581495 15     1203.9 80.26
581496 32     598.97 18.72
581497 59     3217.2 54.53
581498 218    5664.89 25.99
581500 3      103.3  34.43
581501 24     485.0  20.21
581502 20     133.35 6.67
581514 24     200.99 8.37
581516 15     382.01 25.47
581538 74     197.09 2.66
581566 1      469.44 469.44
581567 12     389.21 32.43
581570 13     224.95 17.3
581571 20     494.6  24.73
581572 4      250.52 62.63
581574 13     277.08 21.31
581578 38     848.55 22.33
581579 30     903.62 30.12
581580 24     94.11  3.92
581581 3      51.18  17.06
581582 2      29.88  14.94
```

```
YTRUN-03_average_invoice_value
581582 2      29.88  14.94
581583 2      124.6  62.3
581584 2      140.64 70.32
581585 21     329.05 15.67
581586 4      339.2  84.8
581587 15     249.45 16.63
A563185 1      11062.06 11062.06
ZZZ_AVERAGE_INVOICE_VALUE 19960 1.066668454E7 534.4
```

## Top 10 Customers by Spending

Returns the ten customers with the highest total spend, in descending order. Rows carrying no CustomerID are anonymous till transactions and are excluded.

1. Mapper Class : emits CustomerID of type Text and line revenue of type DoubleWritable
2. Reducer Class : totals each customer and retains only the ten highest in a TreeMap, written from cleanup()
3. Driver Class : sets a single reducer so one ranking is produced

## Code

```
1 package topcustomersbyspending;
2
3 import java.io.IOException;
4 import java.util.Map;
5 import java.util.TreeMap;
6
7 import org.apache.hadoop.conf.Configuration;
8 import org.apache.hadoop.fs.Path;
9 import org.apache.hadoop.io.DoubleWritable;
10 import org.apache.hadoop.io.LongWritable;
11 import org.apache.hadoop.io.Text;
12 import org.apache.hadoop.mapreduce.Job;
13 import org.apache.hadoop.mapreduce.Mapper;
14 import org.apache.hadoop.mapreduce.Reducer;
15 import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
16 import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;
17
18 /**
19  * Top 10 Customers by Spending : returns the ten customers with the highest
20  * total spend, in descending order.
21  *
22  * 135,080 of the 541,909 rows carry no CustomerID; those are anonymous
23  * till transactions and are excluded, since they cannot be attributed.
24  */
```

```
25 public class TopCustomersBySpending {
26
27     public static final int TOP_N = 10;
28
29     public static class Map1 extends Mapper<LongWritable, Text, Text, DoubleWritable> {
30
31         private Text customer = new Text();
32         private DoubleWritable spend = new DoubleWritable();
33
34         @Override
35         public void map(LongWritable key, Text value, Context context)
36             throws IOException, InterruptedException {
37
38             String[] col = value.toString().split(",");
39             if (col.length < 8) {
40                 return;
41             }
42             if (col[0].startsWith("C") || col[0].startsWith("c")) {
43                 context.getCounter("Retail", "CancelledRowsSkipped").increment(1);
44                 return;
45             }
46             if (col[6] == null || col[6].trim().isEmpty()) {
47                 context.getCounter("Retail", "NoCustomerIdSkipped").increment(1);
48                 return;
49             }
50         }
51     }
52 }
```

```

49     }
50
51     try {
52         double qty = Double.parseDouble(col[3]);
53         double price = Double.parseDouble(col[5]);
54         if (qty <= 0 || price <= 0) {
55             return;
56         }
57         customer.set(col[6].trim());
58         spend.set(qty * price);
59         context.write(customer, spend);
60     } catch (NumberFormatException e) {
61         context.getCounter("Retail", "UnparseableRowsSkipped").increment(1);
62     }
63 }
64 }
65
66 public static class Reduce1 extends Reducer<Text, DoubleWritable, Text, DoubleWritable> {
67
68     // Ordered by spend; only the top N are ever retained, so memory stays bounded
69     // no matter how many customers the dataset holds.
70     private TreeMap<Double, String> ranked = new TreeMap<Double, String>();
71
72     @Override

```

```

73     public void reduce(Text key, Iterable<DoubleWritable> values, Context context)
74         throws IOException, InterruptedException {
75
76         double sum = 0;
77         for (DoubleWritable value : values) {
78             sum += value.get();
79         }
80         ranked.put(sum, key.toString());
81         if (ranked.size() > TOP_N) {
82             ranked.remove(ranked.firstKey());
83         }
84     }
85
86     @Override
87     protected void cleanup(Context context) throws IOException, InterruptedException {
88         for (Map.Entry<Double, String> e : ranked.descendingMap().entrySet()) {
89             context.write(new Text(e.getValue()),
90                 new DoubleWritable(Math.round(e.getKey() * 100.0) / 100.0));
91         }
92     }
93 }
94
95 public static void main(String[] args)
96     throws IOException, InterruptedException, ClassNotFoundException {

```

```

97
98     Configuration conf = new Configuration();
99     Job job = Job.getInstance(conf, "Top 10 Customers By Spending");
100    job.setJarByClass(TopCustomersBySpending.class);
101
102    job.setMapperClass(Map1.class);
103    job.setReducerClass(Reduce1.class);
104
105    job.setMapOutputKeyClass(Text.class);
106    job.setMapOutputValueClass(DoubleWritable.class);
107    job.setOutputKeyClass(Text.class);
108    job.setOutputValueClass(DoubleWritable.class);
109
110    // One reducer, so a single ranking is produced rather than one per partition.
111    job.setNumReduceTasks(1);
112
113    FileInputFormat.addInputPath(job, new Path(args[0]));
114    FileOutputFormat.setOutputPath(job, new Path(args[1]));
115
116    System.exit(job.waitForCompletion(true) ? 0 : 1);
117 }
118 }

```

## Execution

```

YTRUN-04_top_customers_by_spending

D:\Desktop\bdal_project>javac -classpath C:\hadoop\share\hadoop\common\*;C:\hadoop\share\hadoop\mapreduce\
*;C:\hadoop\share\hadoop\hdfs\*;C:\hadoop\share\hadoop\common\lib* -d D:\Desktop\bdal_project\TopCustom
sBySpending\target\classes D:\Desktop\bdal_project\TopCustomersBySpending\src\main\java\topcustomersbyspen
ding\TopCustomersBySpending.java

D:\Desktop\bdal_project>jar -cvf D:\Desktop\bdal_project\TopCustomersBySpending\target\TopCustomersBySpen
ding.jar -C D:\Desktop\bdal_project\TopCustomersBySpending\target\classes .
added manifest
adding: topcustomersbyspending/(in = 0) (out= 0)(stored 0%)
adding: topcustomersbyspending/TopCustomersBySpending$Map1.class(in = 2641) (out= 1151)(deflated 56%)
adding: topcustomersbyspending/TopCustomersBySpending$Reduce1.class(in = 3031) (out= 1248)(deflated 58%)
adding: topcustomersbyspending/TopCustomersBySpending.class(in = 1814) (out= 943)(deflated 48%)

D:\Desktop\bdal_project>hadoop jar D:\Desktop\bdal_project\TopCustomersBySpending\target\TopCustomersBySp
ending.jar topcustomersbyspending.TopCustomersBySpending /BdalProject/data /BdalProject/output/topcustomers
2026-09-08 17:48:43,495 INFO client.RMProxy: Connecting to ResourceManager at /0.0.0.0:8032
2026-09-08 17:48:44,961 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not perform
ed. Implement the Tool interface and execute your application with ToolRunner to remedy this.
2026-09-08 17:48:44,982 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop
-yarn/staging/HP/.staging/job_1788866353307_0009
2026-09-08 17:48:45,347 INFO Input.FileInputFormat: Total input files to process : 13
2026-09-08 17:48:45,473 INFO mapreduce.JobSubmitter: number of splits:13
2026-09-08 17:48:45,655 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1788866353307_0009
2026-09-08 17:48:45,658 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-09-08 17:48:45,950 INFO conf.Configuration: resource-types.xml not found
2026-09-08 17:48:45,951 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2026-09-08 17:48:46,045 INFO impl.YarnClientImpl: Submitted application application_1788866353307_0009
2026-09-08 17:48:46,117 INFO mapreduce.Job: The url to track the job: http://DESKTOP-S64ND9J:8088/proxy/ap
plication_1788866353307_0009/
2026-09-08 17:48:46,118 INFO mapreduce.Job: Running job: job_1788866353307_0009
2026-09-08 17:49:03,426 INFO mapreduce.Job: Job job_1788866353307_0009 running in uber mode : false
2026-09-08 17:49:03,430 INFO mapreduce.Job: map 0% reduce 0%
2026-09-08 17:49:22,981 INFO mapreduce.Job: map 8% reduce 0%
2026-09-08 17:49:27,169 INFO mapreduce.Job: map 15% reduce 0%
2026-09-08 17:49:32,384 INFO mapreduce.Job: map 23% reduce 0%
2026-09-08 17:49:33,428 INFO mapreduce.Job: map 31% reduce 0%
2026-09-08 17:49:35,485 INFO mapreduce.Job: map 38% reduce 0%
2026-09-08 17:49:38,572 INFO mapreduce.Job: map 46% reduce 0%
2026-09-08 17:49:41,626 INFO mapreduce.Job: map 54% reduce 0%
2026-09-08 17:49:53,876 INFO mapreduce.Job: map 62% reduce 0%
2026-09-08 17:49:56,950 INFO mapreduce.Job: map 69% reduce 0%
2026-09-08 17:50:01,053 INFO mapreduce.Job: map 77% reduce 0%

```

```
YTRUN-04_top_customers_by_spending

Reduce input records=397884
Reduce output records=10
Spilled Records=795768
Shuffled Maps =13
Failed Shuffles=0
Merged Map outputs=13
GC time elapsed (ms)=2030
CPU time spent (ms)=33239
Physical memory (bytes) snapshot=4511465472
Virtual memory (bytes) snapshot=5971415040
Total committed heap usage (bytes)=3747086336
Peak Map Physical memory (bytes)=340942848
Peak Map Virtual memory (bytes)=450437120
Peak Reduce Physical memory (bytes)=222162944
Peak Reduce Virtual memory (bytes)=309403648

Retail
CancelledRowsSkipped=9288
NoCustomerIdSkipped=134697
Shuffle Errors
BAD_ID=0
CONNECTION=0
IO_ERROR=0
WRONG_LENGTH=0
WRONG_MAP=0
WRONG_REDUCE=0
File Input Format Counters
Bytes Read=45458480
File Output Format Counters
Bytes Written=154

D:\Desktop\bdal_project>hdfs dfs -ls /BdalProject/output/topcustomers
Found 2 items
-rw-r--r--  1 HP supergroup          0 2026-09-08 17:50 /BdalProject/output/topcustomers/_SUCCESS
-rw-r--r--  1 HP supergroup       154 2026-09-08 17:50 /BdalProject/output/topcustomers/part-r-000000

D:\Desktop\bdal_project>hdfs dfs -cat /BdalProject/output/topcustomers/part-r-000000
14646  280206.02
18102  259657.3
17450  194550.79
16446  168472.5
14911  143825.06
12415  124014.53
14156  117379.63
```

```
YTRUN-04_top_customers_by_spending

14156  117379.63
17511  91062.38
16029  81024.84
12346  77183.6
```

## Binning by Country

Splits the transactions into one output file per country using MultipleOutputs. No aggregation is required, only routing, so the reduce phase is switched off and the job is map-only.

1. Driver Class : sets up configuration for the Mapper and registers the named output
2. Mapper Class : uses MultipleOutputs to write each row to the file for its country

## Code

```
1 package binningbycountry;
2
3 import java.io.IOException;
4
5 import org.apache.hadoop.conf.Configuration;
6 import org.apache.hadoop.fs.Path;
7 import org.apache.hadoop.io.LongWritable;
8 import org.apache.hadoop.io.NullWritable;
9 import org.apache.hadoop.io.Text;
10 import org.apache.hadoop.mapreduce.Job;
11 import org.apache.hadoop.mapreduce.Mapper;
12 import org.apache.hadoop.mapreduce.lib.input.TextInputFormat;
13 import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;
14 import org.apache.hadoop.mapreduce.lib.output.MultipleOutputs;
15 import org.apache.hadoop.mapreduce.lib.output.TextOutputFormat;
16
17 /**
18  * Binning by Country : splits the transactions into one output file per
19  * country using MultipleOutputs. This is a map-only job - no aggregation is
20  * required, only routing - so the reduce phase is switched off entirely.
21  */
22 public class BinningByCountry {
23
24     public static class BinMapper extends Mapper<LongWritable, Text, Text, NullWritable> {
```

```
25
26         private MultipleOutputs<Text, NullWritable> mos = null;
27
28         @Override
29         protected void setup(Context context) throws IOException, InterruptedException {
30             mos = new MultipleOutputs<Text, NullWritable>(context);
31         }
32
33         @Override
34         protected void map(LongWritable key, Text value, Context context)
35             throws IOException, InterruptedException {
36
37             String[] col = value.toString().split(",");
38             if (col.length < 8) {
39                 return;
40             }
41             String country = col[7].trim();
42             if (country.isEmpty()) {
43                 country = "Unspecified";
44             }
45             // The bin name becomes part of a file name, so anything that is not a
46             // letter or digit is folded to an underscore.
47             String bin = country.replaceAll("[^A-Za-z0-9]", "_");
48             mos.write("bins", value, NullWritable.get(), bin);
```

```

49         context.getCounter("Country", country).increment(1);
50     }
51
52     @Override
53     protected void cleanup(Context context) throws IOException, InterruptedException {
54         mos.close();
55     }
56 }
57
58 public static void main(String[] args)
59     throws IOException, InterruptedException, ClassNotFoundException {
60
61     Configuration conf = new Configuration();
62     Job job = Job.getInstance(conf, "Binning By Country");
63     job.setJarByClass(BinningByCountry.class);
64     job.setMapperClass(BinMapper.class);
65     job.setNumReduceTasks(0);
66
67     job.setOutputKeyClass(Text.class);
68     job.setOutputValueClass(NullWritable.class);
69
70     TextInputFormat.setInputPaths(job, new Path(args[0]));
71     FileOutputFormat.setOutputPath(job, new Path(args[1]));
72
73     MultipleOutputs.addNamedOutput(job, "bins", TextOutputFormat.class,
74         Text.class, NullWritable.class);
75     MultipleOutputs.setCountersEnabled(job, true);
76
77     System.exit(job.waitForCompletion(true) ? 0 : 1);
78 }
79 }

```

## Execution

```

YTRUN-05_binning_by_country

D:\Desktop\bdal_project>javac -classpath C:\hadoop\share\hadoop\common\*;C:\hadoop\share\hadoop\mapreduce\
*;C:\hadoop\share\hadoop\hdfs\*;C:\hadoop\share\hadoop\common\lib* -d D:\Desktop\bdal_project\BinningByCo
untry\target\classes D:\Desktop\bdal_project\BinningByCountry\src\main\java\binningbycountry\BinningByCoun
try.java

D:\Desktop\bdal_project>jar -cvf D:\Desktop\bdal_project\BinningByCountry\target\BinningByCountry.jar -C D
:\Desktop\bdal_project\BinningByCountry\target\classes .
added manifest
adding: binningbycountry/(in = 0) (out= 0)(stored 0%)
adding: binningbycountry/BinningByCountry$BinMapper.class(in = 2928) (out= 1073)(deflated 63%)
adding: binningbycountry/BinningByCountry.class(in = 1967) (out= 952)(deflated 51%)

D:\Desktop\bdal_project>hadoop jar D:\Desktop\bdal_project\BinningByCountry\target\BinningByCountry.jar bi
nningbycountry.BinningByCountry /BdalProject/data /BdalProject/output/binningbycountry
2026-09-08 17:50:53,328 INFO client.RMProxy: Connecting to ResourceManager at /0.0.0.0:8032
2026-09-08 17:50:54,470 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not perform
ed. Implement the Tool interface and execute your application with ToolRunner to remedy this.
2026-09-08 17:50:54,485 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop
-yarn/staging/HP/.staging/job_1788866353307_0010
2026-09-08 17:50:54,737 INFO Input.FileInputFormat: Total input files to process : 13
2026-09-08 17:50:54,815 INFO mapreduce.JobSubmitter: number of splits:13
2026-09-08 17:50:54,996 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1788866353307_0010
2026-09-08 17:50:54,997 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-09-08 17:50:55,256 INFO conf.Configuration: resource-types.xml not found
2026-09-08 17:50:55,257 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2026-09-08 17:50:55,343 INFO impl.YarnClientImpl: Submitted application application_1788866353307_0010
2026-09-08 17:50:55,394 INFO mapreduce.Job: The url to track the job: http://DESKTOP-S64ND9J:8088/proxy/ap
plication_1788866353307_0010/
2026-09-08 17:50:55,397 INFO mapreduce.Job: Running job: job_1788866353307_0010
2026-09-08 17:51:11,709 INFO mapreduce.Job: Job job_1788866353307_0010 running in uber mode : false
2026-09-08 17:51:11,711 INFO mapreduce.Job: map 0% reduce 0%
2026-09-08 17:51:36,370 INFO mapreduce.Job: map 8% reduce 0%
2026-09-08 17:51:37,392 INFO mapreduce.Job: map 15% reduce 0%
2026-09-08 17:51:45,652 INFO mapreduce.Job: map 23% reduce 0%
2026-09-08 17:51:47,720 INFO mapreduce.Job: map 31% reduce 0%
2026-09-08 17:51:48,732 INFO mapreduce.Job: map 38% reduce 0%
2026-09-08 17:51:50,811 INFO mapreduce.Job: map 46% reduce 0%
2026-09-08 17:51:53,910 INFO mapreduce.Job: map 54% reduce 0%
2026-09-08 17:52:11,356 INFO mapreduce.Job: map 62% reduce 0%
2026-09-08 17:52:15,551 INFO mapreduce.Job: map 69% reduce 0%
2026-09-08 17:52:17,593 INFO mapreduce.Job: map 85% reduce 0%
2026-09-08 17:52:20,647 INFO mapreduce.Job: map 100% reduce 0%

```

| YTRUN-05_binning_by_country |            |                 |       |            |       |                                                   |  |
|-----------------------------|------------|-----------------|-------|------------|-------|---------------------------------------------------|--|
| 08                          | -rw-r--r-- | 1 HP supergroup | 6503  | 2026-09-08 | 17:51 | /BdalProject/output/binningbycountry/Channel_Isla |  |
| nds-m-00000                 | -rw-r--r-- | 1 HP supergroup | 8970  | 2026-09-08 | 17:51 | /BdalProject/output/binningbycountry/Channel_Isla |  |
| nds-m-00001                 | -rw-r--r-- | 1 HP supergroup | 5772  | 2026-09-08 | 17:51 | /BdalProject/output/binningbycountry/Channel_Isla |  |
| nds-m-00002                 | -rw-r--r-- | 1 HP supergroup | 1449  | 2026-09-08 | 17:51 | /BdalProject/output/binningbycountry/Channel_Isla |  |
| nds-m-00003                 | -rw-r--r-- | 1 HP supergroup | 3153  | 2026-09-08 | 17:51 | /BdalProject/output/binningbycountry/Channel_Isla |  |
| nds-m-00005                 | -rw-r--r-- | 1 HP supergroup | 6980  | 2026-09-08 | 17:51 | /BdalProject/output/binningbycountry/Channel_Isla |  |
| nds-m-00006                 | -rw-r--r-- | 1 HP supergroup | 15687 | 2026-09-08 | 17:52 | /BdalProject/output/binningbycountry/Channel_Isla |  |
| nds-m-00007                 | -rw-r--r-- | 1 HP supergroup | 12130 | 2026-09-08 | 17:52 | /BdalProject/output/binningbycountry/Channel_Isla |  |
| nds-m-00008                 | -rw-r--r-- | 1 HP supergroup | 2728  | 2026-09-08 | 17:52 | /BdalProject/output/binningbycountry/Channel_Isla |  |
| nds-m-00009                 | -rw-r--r-- | 1 HP supergroup | 745   | 2026-09-08 | 17:52 | /BdalProject/output/binningbycountry/Channel_Isla |  |
| nds-m-00010                 | -rw-r--r-- | 1 HP supergroup | 873   | 2026-09-08 | 17:52 | /BdalProject/output/binningbycountry/Channel_Isla |  |
| nds-m-00011                 | -rw-r--r-- | 1 HP supergroup | 536   | 2026-09-08 | 17:52 | /BdalProject/output/binningbycountry/Channel_Isla |  |
| nds-m-00012                 | -rw-r--r-- | 1 HP supergroup | 7755  | 2026-09-08 | 17:51 | /BdalProject/output/binningbycountry/Cyprus-m-000 |  |
| 00                          | -rw-r--r-- | 1 HP supergroup | 12781 | 2026-09-08 | 17:51 | /BdalProject/output/binningbycountry/Cyprus-m-000 |  |
| 01                          | -rw-r--r-- | 1 HP supergroup | 78    | 2026-09-08 | 17:51 | /BdalProject/output/binningbycountry/Cyprus-m-000 |  |
| 02                          | -rw-r--r-- | 1 HP supergroup | 6497  | 2026-09-08 | 17:51 | /BdalProject/output/binningbycountry/Cyprus-m-000 |  |
| 03                          | -rw-r--r-- | 1 HP supergroup | 3579  | 2026-09-08 | 17:51 | /BdalProject/output/binningbycountry/Cyprus-m-000 |  |
| 06                          | -rw-r--r-- | 1 HP supergroup | 3778  | 2026-09-08 | 17:52 | /BdalProject/output/binningbycountry/Cyprus-m-000 |  |
| 07                          | -rw-r--r-- | 1 HP supergroup | 1253  | 2026-09-08 | 17:52 | /BdalProject/output/binningbycountry/Cyprus-m-000 |  |
| 09                          | -rw-r--r-- | 1 HP supergroup | 169   | 2026-09-08 | 17:52 | /BdalProject/output/binningbycountry/Cyprus-m-000 |  |
| 10                          | -rw-r--r-- | 1 HP supergroup | 12143 | 2026-09-08 | 17:52 | /BdalProject/output/binningbycountry/Cyprus-m-000 |  |
| 11                          | -rw-r--r-- | 1 HP supergroup |       |            |       |                                                   |  |

| YTRUN-05_binning_by_country |            |                 |        |            |       |                                                   |  |
|-----------------------------|------------|-----------------|--------|------------|-------|---------------------------------------------------|--|
| 010                         | -rw-r--r-- | 1 HP supergroup | 846    | 2026-09-08 | 17:52 | /BdalProject/output/binningbycountry/Finland-m-00 |  |
| 011                         | -rw-r--r-- | 1 HP supergroup | 1031   | 2026-09-08 | 17:52 | /BdalProject/output/binningbycountry/Finland-m-00 |  |
| 012                         | -rw-r--r-- | 1 HP supergroup | 122080 | 2026-09-08 | 17:51 | /BdalProject/output/binningbycountry/France-m-000 |  |
| 00                          | -rw-r--r-- | 1 HP supergroup | 69477  | 2026-09-08 | 17:51 | /BdalProject/output/binningbycountry/France-m-000 |  |
| 01                          | -rw-r--r-- | 1 HP supergroup | 80616  | 2026-09-08 | 17:51 | /BdalProject/output/binningbycountry/France-m-000 |  |
| 02                          | -rw-r--r-- | 1 HP supergroup | 33820  | 2026-09-08 | 17:51 | /BdalProject/output/binningbycountry/France-m-000 |  |
| 03                          | -rw-r--r-- | 1 HP supergroup | 34790  | 2026-09-08 | 17:51 | /BdalProject/output/binningbycountry/France-m-000 |  |
| 04                          | -rw-r--r-- | 1 HP supergroup | 55535  | 2026-09-08 | 17:51 | /BdalProject/output/binningbycountry/France-m-000 |  |
| 05                          | -rw-r--r-- | 1 HP supergroup | 46782  | 2026-09-08 | 17:51 | /BdalProject/output/binningbycountry/France-m-000 |  |
| 06                          | -rw-r--r-- | 1 HP supergroup | 42714  | 2026-09-08 | 17:52 | /BdalProject/output/binningbycountry/France-m-000 |  |
| 07                          | -rw-r--r-- | 1 HP supergroup | 43959  | 2026-09-08 | 17:52 | /BdalProject/output/binningbycountry/France-m-000 |  |
| 08                          | -rw-r--r-- | 1 HP supergroup | 52052  | 2026-09-08 | 17:52 | /BdalProject/output/binningbycountry/France-m-000 |  |
| 09                          | -rw-r--r-- | 1 HP supergroup | 17790  | 2026-09-08 | 17:52 | /BdalProject/output/binningbycountry/France-m-000 |  |
| 10                          | -rw-r--r-- | 1 HP supergroup | 32793  | 2026-09-08 | 17:52 | /BdalProject/output/binningbycountry/France-m-000 |  |
| 11                          | -rw-r--r-- | 1 HP supergroup | 25969  | 2026-09-08 | 17:52 | /BdalProject/output/binningbycountry/France-m-000 |  |
| 12                          | -rw-r--r-- | 1 HP supergroup | 89832  | 2026-09-08 | 17:51 | /BdalProject/output/binningbycountry/Germany-m-00 |  |
| 000                         | -rw-r--r-- | 1 HP supergroup | 103471 | 2026-09-08 | 17:51 | /BdalProject/output/binningbycountry/Germany-m-00 |  |
| 001                         | -rw-r--r-- | 1 HP supergroup | 63104  | 2026-09-08 | 17:51 | /BdalProject/output/binningbycountry/Germany-m-00 |  |
| 002                         | -rw-r--r-- | 1 HP supergroup | 44175  | 2026-09-08 | 17:51 | /BdalProject/output/binningbycountry/Germany-m-00 |  |
| 003                         | -rw-r--r-- | 1 HP supergroup | 61112  | 2026-09-08 | 17:51 | /BdalProject/output/binningbycountry/Germany-m-00 |  |
| 004                         | -rw-r--r-- | 1 HP supergroup | 60833  | 2026-09-08 | 17:51 | /BdalProject/output/binningbycountry/Germany-m-00 |  |
| 005                         | -rw-r--r-- | 1 HP supergroup |        |            |       |                                                   |  |

## Pig Analysis

The Pig scripts were executed in local mode against the same comma separated dataset used by the MapReduce jobs.

### 1) Top 5 Countries by Revenue

Returns the five countries generating the highest total sales value.

#### Script

```
1 -- 1) Top 5 Countries by Revenue : returns the five countries generating the
2 -- highest total sales value.
3 retail = LOAD 'D:\Desktop\bdal_project\data\monthly' USING PigStorage(',') AS
4 (invoiceno:chararray, stockcode:chararray, description:chararray, quantity:int,
5 invoicedate:chararray, unitprice:double, customerid:chararray, country:chararray);
6 -- an InvoiceNo starting with C is a cancellation and would subtract from real sales
7 valid = FILTER retail BY country IS NOT NULL AND quantity > 0 AND unitprice > 0
8 AND NOT (invoiceno MATCHES 'C.*');
9 lines = FOREACH valid GENERATE country, (quantity * unitprice) AS revenue;
10 grp_country = GROUP lines BY country;
11 revenue_by_country = FOREACH grp_country GENERATE group AS country,
12 ROUND(SUM(lines.revenue) * 100) / 100.0 AS total_revenue;
13 sorted_desc = ORDER revenue_by_country BY total_revenue DESC;
14 top5_countries = LIMIT sorted_desc 5;
15 STORE top5_countries INTO 'D:\Desktop\bdal_project\PigAnalysis\output\Top5CountriesByRevenue' USING PigStorage(',');
```

#### Execution

```
YTRUN-06_pig_top5_countries_by_revenue
D:\Desktop\bdal_project> pig -x local D:\Desktop\bdal_project\PigAnalysis\scripts\top5_countries_by_revenue.pig
2026-09-08 17:54:04,466 INFO pig.ExecTypeProvider: Trying ExecType : LOCAL
2026-09-08 17:54:04,467 INFO pig.ExecTypeProvider: Picked LOCAL as the ExecType
2026-09-08 17:54:05,176 [main] INFO org.apache.pig.Main - Apache Pig version 0.17.0 (r1797386) compiled Jun 02
2017, 15:41:58
2026-09-08 17:54:05,177 [main] INFO org.apache.pig.Main - Logging error messages to: C:\hadoop\logs\pig_1788868
445172.log
2026-09-08 17:54:05,207 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - user.name is deprecated.
Instead, use mapreduce.job.user.name
2026-09-08 17:54:05,876 [main] INFO org.apache.pig.impl.util.Utils - Default bootup file C:\Users\HP/.pigbootup
not found
2026-09-08 17:54:05,968 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker is de
precated. Instead, use mapreduce.jobtracker.address
2026-09-08 17:54:05,970 [main] INFO org.apache.pig.backend.hadoop.executionengine.HExecutionEngine - Connecting
to hadoop file system at: file:///
2026-09-08 17:54:06,006 [main] INFO org.apache.pig.PigServer - Pig Script ID for the session: PIG-top5_countrie
s_by_revenue.pig-8659662c-67b9-4e0c-b8ca-1df75cca6579
2026-09-08 17:54:06,006 [main] WARN org.apache.pig.PigServer - ATS is disabled since yarn.timeline-service.enab
led set to false
2026-09-08 17:54:07,051 [main] WARN org.apache.pig.newplan.BaseOperatorPlan - Encountered Warning IMPLICIT_CAST
_TO_DOUBLE 4 time(s).
2026-09-08 17:54:07,056 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.textoutputformat.
separator is deprecated. Instead, use mapreduce.output.textoutputformat.separator
2026-09-08 17:54:07,080 [main] INFO org.apache.pig.tools.pigstats.ScriptState - Pig features used in the script
: GROUP BY, ORDER BY, FILTER, LIMIT
2026-09-08 17:54:07,237 [main] INFO org.apache.pig.newplan.logical.optimizer.LogicalPlanOptimizer - {RULES_ENAB
LED=[AddForEach, ColumnMapKeyPrune, ConstantCalculator, GroupByConstParallelSetter, LimitOptimizer, LoadTypeCast
Insertion, MergeFilter, MergeForEach, NestedLimitOptimizer, PartitionFilterOptimizer, PredicatePushdownOptimizer,
PushDownForEachFlatten, PushUpFilter, SplitFilter, StreamTypeCastInsertion]}
2026-09-08 17:54:07,321 [main] INFO org.apache.pig.impl.util.SpillableMemoryManager - Selected heap (PS Old Gen
) of size 699400192 to monitor. collectionUsageThreshold = 489580128, usageThreshold = 489580128
2026-09-08 17:54:07,391 [main] INFO org.apache.pig.newplan.logical.rules.ColumnPruneVisitor - Columns pruned fo
r retail: $1, $2, $4, $6
2026-09-08 17:54:07,521 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MRCompiler - F
ile concatenation threshold: 100 optimistic? false
2026-09-08 17:54:07,600 [main] INFO org.apache.pig.backend.hadoop.executionengine.util.CombinerOptimizerUtil -
Choosing to move algebraic foreach to combiner
2026-09-08 17:54:07,624 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.SecondaryKeyOp
timizerMR - Using Secondary Key Optimization for MapReduce node scope-67
2026-09-08 17:54:07,634 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOpti
mizer - MR plan size before optimization: 4
2026-09-08 17:54:07,635 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOpti
```

```
YTRUN-06_pig_top5_countries_by_revenue
2026-09-08 17:54:05,970 [main] INFO org.apache.pig.backend.hadoop.executionengine.HExecutionEngine - Connecting
to hadoop file system at: file:///
2026-09-08 17:54:06,006 [main] INFO org.apache.pig.PigServer - Pig Script ID for the session: PIG-top5_countrie
s_by_revenue.pig-8659662c-67b9-4e0c-b8ca-1df75cca6579
2026-09-08 17:54:06,006 [main] WARN org.apache.pig.PigServer - ATS is disabled since yarn.timeline-service.enab
led set to false
2026-09-08 17:54:07,051 [main] WARN org.apache.pig.newplan.BaseOperatorPlan - Encountered Warning IMPLICIT_CAST
_TO_DOUBLE 4 time(s).
2026-09-08 17:54:07,056 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.textoutputformat.
separator is deprecated. Instead, use mapreduce.output.textoutputformat.separator
2026-09-08 17:54:07,080 [main] INFO org.apache.pig.tools.pigstats.ScriptState - Pig features used in the script
: GROUP_BY,ORDER_BY,FILTER,LIMIT
2026-09-08 17:54:07,237 [main] INFO org.apache.pig.newplan.logical.optimizer.LogicalPlanOptimizer - {RULES_ENAB
LED=[AddForEach, ColumnMapKeyPrune, ConstantCalculator, GroupByConstParallelSetter, LimitOptimizer, LoadTypeCast
Inserter, MergeFilter, MergeForEach, NestedLimitOptimizer, PartitionFilterOptimizer, PredicatePushdownOptimizer,
PushDownForEachFlatten, PushUpFilter, SplitFilter, StreamTypeCastInserter]}
2026-09-08 17:54:07,321 [main] INFO org.apache.pig.impl.util.SpillableMemoryManager - Selected heap (PS Old Gen
) of size 699400192 to monitor. collectionUsageThreshold = 489580128, usageThreshold = 489580128
2026-09-08 17:54:07,391 [main] INFO org.apache.pig.newplan.logical.rules.ColumnPruneVisitor - Columns pruned fo
r retail: $1, $2, $4, $6
2026-09-08 17:54:07,521 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MRCompiler - F
ile concatenation threshold: 100 optimistic? false
2026-09-08 17:54:07,600 [main] INFO org.apache.pig.backend.hadoop.executionengine.util.CombinerOptimizerUtil -
Choosing to move algebraic foreach to combiner
2026-09-08 17:54:07,624 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.SecondaryKeyOp
timizerMR - Using Secondary Key Optimization for MapReduce node scope-67
2026-09-08 17:54:07,634 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOpti
mizer - MR plan size before optimization: 4
2026-09-08 17:54:07,635 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOpti
mizer - MR plan size after optimization: 4
2026-09-08 17:54:07,743 [main] INFO org.apache.hadoop.metrics2.impl.MetricsConfig - Loaded properties from had
op-metrics2.properties
2026-09-08 17:54:07,849 [main] INFO org.apache.hadoop.metrics2.impl.MetricsSystemImpl - Scheduled Metric snapsh
ot period at 10 second(s).
2026-09-08 17:54:07,850 [main] INFO org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics syst
em started
2026-09-08 17:54:07,888 [main] INFO org.apache.pig.tools.pigstats.mapreduce.MRScriptState - Pig script settings
are added to the job
2026-09-08 17:54:07,904 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.reduce.markre
set.buffer.percent is deprecated. Instead, use mapreduce.reduce.markreset.buffer.percent
2026-09-08 17:54:07,905 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlComp
iler - mapred.job.reduce.markreset.buffer.percent is not set, set to default 0.3
2026-09-08 17:54:07,907 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.output.compress i
```

```
YTRUN-06_pig_top5_countries_by_revenue
D:\Desktop\bdal_project>pig -x local D:\Desktop\bdal_project\PigAnalysis\scripts\top5_countries_by_revenue.pig
2026-09-08 17:54:04,466 INFO pig.ExecTypeProvider: Trying ExecType : LOCAL
2026-09-08 17:54:04,467 INFO pig.ExecTypeProvider: Picked LOCAL as the ExecType
2026-09-08 17:54:05,176 [main] INFO org.apache.pig.Main - Apache Pig version 0.17.0 (r1797386) compiled Jun 02
2017, 15:41:58
2026-09-08 17:54:05,177 [main] INFO org.apache.pig.Main - Logging error messages to: C:\hadoop\logs\pig-1788868
445172.log
2026-09-08 17:54:05,207 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - user.name is deprecated.
Instead, use mapreduce.job.user.name
2026-09-08 17:54:05,876 [main] INFO org.apache.pig.impl.util.Utils - Default bootstrap file C:\Users\HP/.pigbootup
not found
2026-09-08 17:54:05,968 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker is de
precated. Instead, use mapreduce.jobtracker.address
2026-09-08 17:54:05,970 [main] INFO org.apache.pig.backend.hadoop.executionengine.HExecutionEngine - Connecting
to hadoop file system at: file:///
2026-09-08 17:54:06,006 [main] INFO org.apache.pig.PigServer - Pig Script ID for the session: PIG-top5_countrie
s_by_revenue.pig-8659662c-67b9-4e0c-b8ca-1df75cca6579
2026-09-08 17:54:06,006 [main] WARN org.apache.pig.PigServer - ATS is disabled since yarn.timeline-service.enab
led set to false
2026-09-08 17:54:07,051 [main] WARN org.apache.pig.newplan.BaseOperatorPlan - Encountered Warning IMPLICIT_CAST
_TO_DOUBLE 4 time(s).
2026-09-08 17:54:07,056 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.textoutputformat.
separator is deprecated. Instead, use mapreduce.output.textoutputformat.separator
2026-09-08 17:54:07,080 [main] INFO org.apache.pig.tools.pigstats.ScriptState - Pig features used in the script
: GROUP_BY,ORDER_BY,FILTER,LIMIT
2026-09-08 17:54:07,237 [main] INFO org.apache.pig.newplan.logical.optimizer.LogicalPlanOptimizer - {RULES_ENAB
LED=[AddForEach, ColumnMapKeyPrune, ConstantCalculator, GroupByConstParallelSetter, LimitOptimizer, LoadTypeCast
Inserter, MergeFilter, MergeForEach, NestedLimitOptimizer, PartitionFilterOptimizer, PredicatePushdownOptimizer,
PushDownForEachFlatten, PushUpFilter, SplitFilter, StreamTypeCastInserter]}
2026-09-08 17:54:07,321 [main] INFO org.apache.pig.impl.util.SpillableMemoryManager - Selected heap (PS Old Gen
) of size 699400192 to monitor. collectionUsageThreshold = 489580128, usageThreshold = 489580128
2026-09-08 17:54:07,391 [main] INFO org.apache.pig.newplan.logical.rules.ColumnPruneVisitor - Columns pruned fo
r retail: $1, $2, $4, $6
2026-09-08 17:54:07,521 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MRCompiler - F
ile concatenation threshold: 100 optimistic? false
2026-09-08 17:54:07,600 [main] INFO org.apache.pig.backend.hadoop.executionengine.util.CombinerOptimizerUtil -
Choosing to move algebraic foreach to combiner
2026-09-08 17:54:07,624 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.SecondaryKeyOp
timizerMR - Using Secondary Key Optimization for MapReduce node scope-67
2026-09-08 17:54:07,634 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOpti
mizer - MR plan size before optimization: 4
2026-09-08 17:54:07,635 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOpti
```

## 2) Top 10 Products by Revenue

Returns the ten products earning the most revenue.

### Script

```
1  -- 2) Top 10 Products by Revenue : returns the ten products earning the most.
2  retail = LOAD 'D:/Desktop/bdal_project/data/monthly' USING PigStorage(',') AS
3      (invoiceno:chararray, stockcode:chararray, description:chararray, quantity:int,
4      invoicedate:chararray, unitprice:double, customerid:chararray, country:chararray);
5  valid = FILTER retail BY quantity > 0 AND unitprice > 0
6      AND NOT (invoiceno MATCHES 'C.*');
7  lines = FOREACH valid GENERATE stockcode, description, (quantity * unitprice) AS revenue;
8  grp_product = GROUP lines BY stockcode;
9  revenue_by_product = FOREACH grp_product GENERATE group AS stockcode,
10      MAX(lines.description) AS description,
11      ROUND(SUM(lines.revenue) * 100) / 100.0 AS total_revenue;
12  sorted_desc = ORDER revenue_by_product BY total_revenue DESC;
13  top10_products = LIMIT sorted_desc 10;
14  STORE top10_products INTO 'D:/Desktop/bdal_project/PigAnalysis/output/Top10ProductsByRevenue' USING PigStorage('|');
```

### Execution

```
YTRUN-07_pig_top10_products_by_revenue
D:\Desktop\bdal_project>pig -x local D:\Desktop\bdal_project\PigAnalysis\scripts\top10_products_by_revenue.pig
2026-09-08 17:54:47,300 INFO pig.ExecTypeProvider: Trying ExecType : LOCAL
2026-09-08 17:54:47,302 INFO pig.ExecTypeProvider: Picked LOCAL as the ExecType
2026-09-08 17:54:48,053 [main] INFO org.apache.pig.Main - Apache Pig version 0.17.0 (r1797386) compiled Jun 02
2017, 15:41:58
2026-09-08 17:54:48,053 [main] INFO org.apache.pig.Main - Logging error messages to: C:\hadoop\logs\pig_1788868
488047.log
2026-09-08 17:54:48,089 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - user.name is deprecated.
Instead, use mapreduce.job.user.name
2026-09-08 17:54:49,157 [main] INFO org.apache.pig.impl.util.Utils - Default bootup file C:\Users\HP/.pigbootup
not found
2026-09-08 17:54:49,368 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker is de
precated. Instead, use mapreduce.jobtracker.address
2026-09-08 17:54:49,374 [main] INFO org.apache.pig.backend.hadoop.executionengine.HExecutionEngine - Connecting
to hadoop file system at: file:///
2026-09-08 17:54:49,491 [main] INFO org.apache.pig.PigServer - Pig Script ID for the session: PIG-top10_product
s_by_revenue.pig-391161be-b787-492f-afff-5d0df2f029fe
2026-09-08 17:54:49,497 [main] WARN org.apache.pig.PigServer - ATS is disabled since yarn.timeline-service.enab
led set to false
2026-09-08 17:54:50,546 [main] WARN org.apache.pig.newplan.BaseOperatorPlan - Encountered Warning IMPLICIT_CAST
_TO_DOUBLE 4 time(s).
2026-09-08 17:54:50,552 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.textoutputformat.
separator is deprecated. Instead, use mapreduce.output.textoutputformat.separator
2026-09-08 17:54:50,585 [main] INFO org.apache.pig.tools.pigstats.ScriptState - Pig features used in the script
: GROUP_BY, ORDER_BY, FILTER, LIMIT
2026-09-08 17:54:50,764 [main] INFO org.apache.pig.newplan.logical.optimizer.LogicalPlanOptimizer - {RULES_ENAB
LED=[AddForEach, ColumnMapKeyPrune, ConstantCalculator, GroupByConstParallelSetter, LimitOptimizer, LoadTypeCast
Inserter, MergeFilter, MergeForEach, NestedLimitOptimizer, PartitionFilterOptimizer, PredicatePushdownOptimizer,
PushDownForEachFlatten, PushUpFilter, SplitFilter, StreamTypeCastInserter]}
2026-09-08 17:54:50,840 [main] INFO org.apache.pig.impl.util.SpillableMemoryManager - Selected heap (PS Old Gen
) of size 699400192 to monitor. collectionUsageThreshold = 489580128, usageThreshold = 489580128
2026-09-08 17:54:50,924 [main] INFO org.apache.pig.newplan.logical.rules.ColumnPruneVisitor - Columns pruned fo
r retail: $4, $6, $7
2026-09-08 17:54:51,077 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MRCompiler - F
ile concatenation threshold: 100 optimistic? false
2026-09-08 17:54:51,151 [main] INFO org.apache.pig.backend.hadoop.executionengine.util.CombinerOptimizerUtil -
Choosing to move algebraic foreach to combiner
2026-09-08 17:54:51,190 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.SecondaryKeyOp
timizerMR - Using Secondary Key Optimization for MapReduce node scope-72
2026-09-08 17:54:51,215 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOpti
mizer - MR plan size before optimization: 4
2026-09-08 17:54:51,215 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOpti
```

```
YTRUN-07_pig_top10_products_by_revenue

WRONG REDUCE=0
File Output Format Counters
Bytes Written=0
2026-09-08 17:55:03,835 [pool-12-thread-1] INFO org.apache.hadoop.mapred.LocalJobRunner - Finishing task: attempt_local1620493607_0003_r_000000_0
2026-09-08 17:55:03,836 [Thread-25] INFO org.apache.hadoop.mapred.LocalJobRunner - reduce task executor complete.
2026-09-08 17:55:03,845 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - HadoopJobId: job_local1620493607_0003
2026-09-08 17:55:03,847 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Processing aliases sorted_desc
2026-09-08 17:55:03,849 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - detailed locations: M: sorted_desc[12,14] C: R:
2026-09-08 17:55:03,855 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - 75% complete
2026-09-08 17:55:03,858 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Running jobs are [job_local1620493607_0003]
2026-09-08 17:55:04,065 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-08 17:55:04,067 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-08 17:55:04,069 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-08 17:55:04,074 [main] INFO org.apache.pig.tools.pigstats.mapreduce.MRScriptState - Pig script settings are added to the job
2026-09-08 17:55:04,074 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - mapred.job.reduce.markreset.buffer.percent is not set, set to default 0.3
2026-09-08 17:55:04,075 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Reduce phase detected, estimating # of required reducers.
2026-09-08 17:55:04,076 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Setting Parallelism to 1
2026-09-08 17:55:04,078 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Setting up single store job
2026-09-08 17:55:04,090 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - 1 map-reduce job(s) waiting for submission.
2026-09-08 17:55:04,092 [JobControl] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-08 17:55:04,104 [JobControl] WARN org.apache.hadoop.mapreduce.JobResourceUploader - No job jar file set. User classes may not be found. See Job or Job#setJar(String).
2026-09-08 17:55:04,117 [JobControl] INFO org.apache.hadoop.mapreduce.lib.input.FileInputFormat - Total input files to process : 1
2026-09-08 17:55:04,117 [JobControl] INFO org.apache.pig.backend.hadoop.executionengine.util.MapRedUtil - Total input paths to process : 1
```

```
YTRUN-07_pig_top10_products_by_revenue

em already initialized!
2026-09-08 17:55:05,099 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Success!
2026-09-08 17:55:05,145 [main] INFO org.apache.pig.Main - Pig script completed in 18 seconds and 91 milliseconds (18091 ms)

D:\Desktop\bda1_project>type D:\Desktop\bda1_project\PigAnalysis\output\Top10ProductsByRevenue\part-r-00000
DOT|DOTCOM POSTAGE|206248.77
22423|REGENCY CAKESTAND 3 TIER|174484.74
23843|PAPER CRAFT LITTLE BIRDIE|168469.6
85123A|WHITE HANGING HEART T-LIGHT HOLDER|104518.8
47566|PARTY BUNTING|99504.33
85099B|JUMBO BAG RED RETROSPOT|94340.05
23166|MEDIUM CERAMIC TOP STORAGE JAR|81700.92
M|Manual|78110.27
POST|POSTAGE|78101.88
23084|RABBIT NIGHT LIGHT|66964.99
```

### 3) Top 10 Products by Country

Returns, for every country, its ten best selling products by revenue, using a nested FOREACH so the ordering and limiting happen inside each group.

## Script

```
1  -- 3) Top 10 Products by Country : returns, for every country, its ten
2  -- best-selling products by revenue. Uses a nested FOREACH so the ordering
3  -- and limiting happen inside each country's group.
4  retail = LOAD 'D:/Desktop/bdal_project/data/monthly' USING PigStorage(',') AS
5  (invoiceno:chararray, stockcode:chararray, description:chararray, quantity:int,
6   invoicedate:chararray, unitprice:double, customerid:chararray, country:chararray);
7  valid = FILTER retail BY country IS NOT NULL AND quantity > 0 AND unitprice > 0
8         AND NOT (invoiceno MATCHES 'C.*');
9  lines = FOREACH valid GENERATE country, stockcode, (quantity * unitprice) AS revenue;
10 grp_cp = GROUP lines BY (country, stockcode);
11 product_totals = FOREACH grp_cp GENERATE group.country AS country,
12         group.stockcode AS stockcode,
13         ROUND(SUM(lines.revenue) * 100) / 100.0 AS total_revenue;
14 grp_country = GROUP product_totals BY country;
15 top10_per_country = FOREACH grp_country {
16         sorted = ORDER product_totals BY total_revenue DESC;
17         top10 = LIMIT sorted 10;
18         GENERATE FLATTEN(top10);
19     };
20 STORE top10_per_country INTO 'D:/Desktop/bdal_project/PigAnalysis/output/Top10ProductsByCountry' USING PigStorage('|');
```

## Execution

```
YTRUN-08_pig_top10_products_by_country
D:\Desktop\bdal_project>pig -x local D:\Desktop\bdal_project\PigAnalysis\scripts\top10_products_by_country.pig
2026-09-08 17:55:52,258 INFO pig.ExecTypeProvider: Trying ExecType : LOCAL
2026-09-08 17:55:52,258 INFO pig.ExecTypeProvider: Picked LOCAL as the ExecType
2026-09-08 17:55:52,912 [main] INFO org.apache.pig.Main - Apache Pig version 0.17.0 (r1797386) compiled Jun 02
2017, 15:41:58
2026-09-08 17:55:52,912 [main] INFO org.apache.pig.Main - Logging error messages to: C:\hadoop\logs\pig_178868
552906.log
2026-09-08 17:55:52,933 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - user.name is deprecated.
Instead, use mapreduce.job.user.name
2026-09-08 17:55:53,522 [main] INFO org.apache.pig.impl.util.Utils - Default bootup file C:\Users\HP/.pigbootup
not found
2026-09-08 17:55:53,589 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker is de
precated. Instead, use mapreduce.jobtracker.address
2026-09-08 17:55:53,591 [main] INFO org.apache.pig.backend.hadoop.executionengine.HExecutionEngine - Connecting
to hadoop file system at: file:///
2026-09-08 17:55:53,627 [main] INFO org.apache.pig.PigServer - Pig Script ID for the session: PIG-top10_product
s_by_country.pig-0ea49196-15e6-4d20-a87c-12a782cb361d
2026-09-08 17:55:53,627 [main] WARN org.apache.pig.PigServer - ATS is disabled since yarn.timeline-service.enab
led set to false
2026-09-08 17:55:54,303 [main] WARN org.apache.pig.newplan.BaseOperatorPlan - Encountered Warning IMPLICIT_CAST
_TO_DOUBLE 4 time(s).
2026-09-08 17:55:54,306 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.textoutputformat.
separator is deprecated. Instead, use mapreduce.output.textoutputformat.separator
2026-09-08 17:55:54,325 [main] INFO org.apache.pig.tools.pigstats.ScriptState - Pig features used in the script
: GROUP_BY,FILTER
2026-09-08 17:55:54,430 [main] INFO org.apache.pig.newplan.logical.optimizer.LogicalPlanOptimizer - {RULES_ENAB
LED=[AddForEach, ColumnMapKeyPrune, ConstantCalculator, GroupByConstParallelSetter, LimitOptimizer, LoadTypeCast
Inserter, MergeFilter, MergeForEach, NestedLimitOptimizer, PartitionFilterOptimizer, PredicatePushdownOptimizer,
PushDownForEachFlatten, PushUpFilter, SplitFilter, StreamTypeCastInserter]}
2026-09-08 17:55:54,470 [main] INFO org.apache.pig.impl.util.SpillableMemoryManager - Selected heap (PS Old Gen
) of size 699400192 to monitor. collectionUsageThreshold = 489580128, usageThreshold = 489580128
2026-09-08 17:55:54,524 [main] INFO org.apache.pig.newplan.logical.rules.ColumnPruneVisitor - Columns pruned fo
r retail: $2, $4, $6
2026-09-08 17:55:54,596 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MRCompiler - F
ile concatenation threshold: 100 optimistic? false
2026-09-08 17:55:54,645 [main] INFO org.apache.pig.backend.hadoop.executionengine.util.CombinerOptimizerUtil -
Choosing to move algebraic foreach to combiner
2026-09-08 17:55:54,670 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOpti
mizer - MR plan size before optimization: 2
2026-09-08 17:55:54,670 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOpti
mizer - MR plan size after optimization: 2
2026-09-08 17:55:54,769 [main] INFO org.apache.hadoop.metrics2.impl.MetricsConfig - Loaded properties from hadoo
```

|        |        |         |
|--------|--------|---------|
| Canada | 22382  | 40.5    |
| Canada | 21558  | 45.9    |
| Canada | 21871  | 45.0    |
| Canada | 23192  | 39.6    |
| Canada | 23190  | 39.6    |
| Cyprus | 22827  | 580.0   |
| Cyprus | 15056N | 392.7   |
| Cyprus | 85123A | 386.4   |
| Cyprus | 22423  | 382.5   |
| Cyprus | M 320  | .69     |
| Cyprus | 22656  | 250.0   |
| Cyprus | 22655  | 250.0   |
| Cyprus | 22335  | 220.8   |
| Cyprus | 22826  | 170.0   |
| Cyprus | 22804  | 152.4   |
| France | POST   | 15454.0 |
| France | M 9492 | .37     |
| France | 23084  | 7277.2  |
| France | 22423  | 2816.85 |
| France | 21731  | 2169.75 |
| France | 22554  | 1868.4  |
| France | 22556  | 1868.4  |
| France | 84879  | 1842.76 |
| France | 22326  | 1837.8  |
| France | 20750  | 1662.9  |
| Greece | POST   | 335.0   |
| Greece | 22423  | 175.2   |
| Greece | 72760B | 135.84  |
| Greece | 48188  | 135.0   |
| Greece | 48184  | 135.0   |
| Greece | 48187  | 135.0   |
| Greece | 48129  | 135.0   |
| Greece | 22692  | 135.0   |
| Greece | 22847  | 119.6   |
| Greece | 22762  | 102.0   |
| Israel | 22423  | 551.1   |
| Israel | 23240  | 254.7   |
| Israel | 22326  | 244.8   |
| Israel | 22192  | 183.6   |
| Israel | 23236  | 159.98  |
| Israel | 21578  | 135.0   |
| Israel | 21246  | 131.7   |
| Israel | 22929  | 130.0   |

|                                        |
|----------------------------------------|
| YTRUN-08_pig_top10_products_by_country |
| -                                      |

#### 4) Top 10 Customers by Spending

Returns the ten customers who spent the most, excluding rows without a CustomerID.

#### Script

```

1  -- 4) Top 10 Customers by Spending : returns the ten customers who spent most.
2  -- Rows without a CustomerID are anonymous till sales and cannot be attributed.
3  retail = LOAD 'D:/Desktop/bdal_project/data/monthly' USING PigStorage(',') AS
4  (invoiceno:chararray, stockcode:chararray, description:chararray, quantity:int,
5   invoicedate:chararray, unitprice:double, customerid:chararray, country:chararray);
6  valid = FILTER retail BY customerid IS NOT NULL AND customerid != ''
7           AND quantity > 0 AND unitprice > 0
8           AND NOT (invoiceno MATCHES 'C.*');
9  lines = FOREACH valid GENERATE customerid, country, (quantity * unitprice) AS revenue;
10 grp_customer = GROUP lines BY customerid;
11 spend_by_customer = FOREACH grp_customer GENERATE group AS customerid,
12                  MAX(lines.country) AS country,
13                  ROUND(SUM(lines.revenue) * 100) / 100.0 AS total_spend;
14 sorted_desc = ORDER spend_by_customer BY total_spend DESC;
15 top10_customers = LIMIT sorted_desc 10;
16 STORE top10_customers INTO 'D:/Desktop/bdal_project/PigAnalysis/output/Top10CustomersBySpending' USING PigStorage('|');

```

## Execution

```
YTRUN-09_pig_top10_customers_by_spending

D:\Desktop\bda1_project>pig -x local D:\Desktop\bda1_project\PigAnalysis\scripts\top10_customers_by_spending.pig
2026-09-08 17:56:49,120 INFO pig.ExecTypeProvider: Trying ExecType : LOCAL
2026-09-08 17:56:49,121 INFO pig.ExecTypeProvider: Picked LOCAL as the ExecType
2026-09-08 17:56:49,784 [main] INFO org.apache.pig.Main - Apache Pig version 0.17.0 (r1797386) compiled Jun 02
2017, 15:41:58
2026-09-08 17:56:49,784 [main] INFO org.apache.pig.Main - Logging error messages to: C:\hadoop\logs\pig_1788868
609780.log
2026-09-08 17:56:49,808 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - user.name is deprecated.
Instead, use mapreduce.job.user.name
2026-09-08 17:56:50,559 [main] INFO org.apache.pig.impl.util.Utils - Default bootup file C:\Users\HP/.pigbootup
not found
2026-09-08 17:56:50,624 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker is de
precated. Instead, use mapreduce.jobtracker.address
2026-09-08 17:56:50,627 [main] INFO org.apache.pig.backend.hadoop.executionengine.HExecutionEngine - Connecting
to hadoop file system at: file:///
2026-09-08 17:56:50,660 [main] INFO org.apache.pig.PigServer - Pig Script ID for the session: PIG-top10_custom
rs_by_spending.pig-9ec207a4-bf4d-4d07-a9f6-20756e3cdeec
2026-09-08 17:56:50,661 [main] WARN org.apache.pig.PigServer - ATS is disabled since yarn.timeline-service.enab
led set to false
2026-09-08 17:56:51,341 [main] WARN org.apache.pig.newplan.BaseOperatorPlan - Encountered Warning IMPLICIT_CAST
_TO_DOUBLE 4 time(s).
2026-09-08 17:56:51,344 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.textoutputformat.
separator is deprecated. Instead, use mapreduce.output.textoutputformat.separator
2026-09-08 17:56:51,365 [main] INFO org.apache.pig.tools.pigstats.ScriptState - Pig features used in the script
: GROUP_BY, ORDER_BY, FILTER, LIMIT
2026-09-08 17:56:51,477 [main] INFO org.apache.pig.newplan.logical.optimizer.LogicalPlanOptimizer - {RULES_ENAB
LED=[AddForEach, ColumnMapKeyPrune, ConstantCalculator, GroupByConstParallelSetter, LimitOptimizer, LoadTypeCast
Inserter, MergeFilter, MergeForEach, NestedLimitOptimizer, PartitionFilterOptimizer, PredicatePushdownOptimizer,
PushDownForEachFlatten, PushUpFilter, SplitFilter, StreamTypeCastInserter]}
2026-09-08 17:56:51,518 [main] INFO org.apache.pig.impl.util.SpillableMemoryManager - Selected heap (PS Old Gen
) of size 699400192 to monitor. collectionUsageThreshold = 489580128, usageThreshold = 489580128
2026-09-08 17:56:51,581 [main] INFO org.apache.pig.newplan.logical.rules.ColumnPruneVisitor - Columns pruned fo
r retail: $1, $2, $4
2026-09-08 17:56:51,672 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MRCompiler - F
ile concatenation threshold: 100 optimistic? false
2026-09-08 17:56:51,726 [main] INFO org.apache.pig.backend.hadoop.executionengine.util.CombinerOptimizerUtil -
Choosing to move algebraic foreach to combiner
2026-09-08 17:56:51,748 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.SecondaryKeyOp
timizerMR - Using Secondary Key Optimization for MapReduce node scope-80
2026-09-08 17:56:51,758 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOpti
mizer - MR plan size before optimization: 4
2026-09-08 17:56:51,758 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOpti
```

```
YTRUN-09_pig_top10_customers_by_spending

Spilled Records=10
Shuffled Maps =1
Failed Shuffles=0
Merged Map outputs=1
GC time elapsed (ms)=0
Total committed heap usage (bytes)=645922816

Shuffle Errors
BAD_ID=0
CONNECTION=0
IO_ERROR=0
WRONG_LENGTH=0
WRONG_MAP=0
WRONG_REDUCE=0

File Output Format Counters
Bytes Written=0
2026-09-08 17:57:00,775 [pool-12-thread-1] INFO org.apache.hadoop.mapred.LocalJobRunner - Finishing task: attem
pt_local1781930435_0003_r_000000_0
2026-09-08 17:57:00,775 [Thread-25] INFO org.apache.hadoop.mapred.LocalJobRunner - reduce task executor complet
e.
2026-09-08 17:57:00,999 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLaunc
her - 75% complete
2026-09-08 17:57:01,002 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics syst
em already initialized!
2026-09-08 17:57:01,004 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics syst
em already initialized!
2026-09-08 17:57:01,006 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics syst
em already initialized!
2026-09-08 17:57:01,010 [main] INFO org.apache.pig.tools.pigstats.mapreduce.MRScriptState - Pig script settings
are added to the job
2026-09-08 17:57:01,011 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlComp
iler - mapred.job.reduce.markreset.buffer.percent is not set, set to default 0.3
2026-09-08 17:57:01,012 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlComp
iler - Reduce phase detected, estimating # of required reducers.
2026-09-08 17:57:01,012 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlComp
iler - Setting Parallelism to 1
2026-09-08 17:57:01,015 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlComp
iler - Setting up single store job
2026-09-08 17:57:01,031 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLaunc
her - 1 map-reduce job(s) waiting for submission.
2026-09-08 17:57:01,043 [JobControl] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metric
s system already initialized!
2026-09-08 17:57:01,063 [JobControl] WARN org.apache.hadoop.mapreduce.JobResourceUploader - No job jar file set
. User classes may not be found. See Job or Job#setJar(String).
```

```
YTRUN-09_pig_top10_customers_by_spending
em already initialized!
2026-09-08 17:57:01,929 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics syst
em already initialized!
2026-09-08 17:57:01,930 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics syst
em already initialized!
2026-09-08 17:57:01,933 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLaunc
her - Success!
2026-09-08 17:57:01,971 [main] INFO org.apache.pig.Main - Pig script completed in 13 seconds and 165 millisecon
ds (13165 ms)

D:\Desktop\bda1_project>type D:\Desktop\bda1_project\PigAnalysis\output\Top10CustomersBySpending\part-r-00000
14646|Netherlands|280206.02
18102|United Kingdom|259657.3
17450|United Kingdom|194550.79
16446|United Kingdom|168472.5
14911|EIRE|143825.06
12415|Australia|124914.53
14156|EIRE|117379.63
17511|United Kingdom|91062.38
16029|United Kingdom|81024.84
12346|United Kingdom|77183.6
```

## 5) Monthly Sales Trend

Returns the total revenue, the number of order lines and the number of distinct invoices for each month of trading.

### Script

```
1  -- 5) Monthly Sales Trend : returns total revenue, order-line count and number
2  --   of distinct invoices for each month of trading.
3  retail = LOAD 'D:/Desktop/bda1_project/data/monthly' USING PigStorage(',') AS
4  (invoiceno:chararray, stockcode:chararray, description:chararray, quantity:int,
5   invoicedate:chararray, unitprice:double, customerid:chararray, country:chararray);
6  valid = FILTER retail BY quantity > 0 AND unitprice > 0
7         AND NOT (invoiceno MATCHES 'C.*');
8  -- InvoiceDate is 'YYYY-MM-DD HH:MM', so the first seven characters are the month
9  lines = FOREACH valid GENERATE SUBSTRING(invoicedate, 0, 7) AS month,
10         invoiceno,
11         (quantity * unitprice) AS revenue;
12  grp_month = GROUP lines BY month;
13  monthly = FOREACH grp_month {
14         invoices = DISTINCT lines.invoiceno;
15         GENERATE group AS month,
16                 ROUND(SUM(lines.revenue) * 100) / 100.0 AS total_revenue,
17                 COUNT(lines) AS order_lines,
18                 COUNT(invoices) AS invoice_count;
19  };
20  monthly_sorted = ORDER monthly BY month ASC;
21  STORE monthly_sorted INTO 'D:/Desktop/bda1_project/PigAnalysis/output/MonthlySalesTrend' USING PigStorage('|');
```

## Execution

```
YTRUN-10_pig_monthly_sales_trend

D:\Desktop\bda1 project>pig -x local D:\Desktop\bda1 project\PigAnalysis\scripts\monthly_sales_trend.pig
2026-09-08 17:57:43,965 INFO pig.ExecTypeProvider: Trying ExecType : LOCAL
2026-09-08 17:57:43,966 INFO pig.ExecTypeProvider: Picked LOCAL as the ExecType
2026-09-08 17:57:44,597 [main] INFO org.apache.pig.Main - Apache Pig version 0.17.0 (r1797386) compiled Jun 02
2017, 15:41:58
2026-09-08 17:57:44,597 [main] INFO org.apache.pig.Main - Logging error messages to: C:\hadoop\logs\pig_1788868
664592.log
2026-09-08 17:57:44,621 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - user.name is deprecated.
Instead, use mapreduce.job.user.name
2026-09-08 17:57:45,296 [main] INFO org.apache.pig.impl.util.Utils - Default bootup file C:\Users\HP/.pigbootup
not found
2026-09-08 17:57:45,372 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker is de
precated. Instead, use mapreduce.jobtracker.address
2026-09-08 17:57:45,374 [main] INFO org.apache.pig.backend.hadoop.executionengine.HExecutionEngine - Connecting
to hadoop file system at: file:///
2026-09-08 17:57:45,417 [main] INFO org.apache.pig.PigServer - Pig Script ID for the session: PIG-monthly_sales
_trend.pig-a0c243d1-e2d8-40d0-a636-cf88dce2b935
2026-09-08 17:57:45,417 [main] WARN org.apache.pig.PigServer - ATS is disabled since yarn.timeline-service.enab
led set to false
2026-09-08 17:57:46,152 [main] WARN org.apache.pig.newplan.BaseOperatorPlan - Encountered Warning IMPLICIT_CAST
_TO_DOUBLE 4 time(s).
2026-09-08 17:57:46,158 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.textoutputformat.
separator is deprecated. Instead, use mapreduce.output.textoutputformat.separator
2026-09-08 17:57:46,188 [main] INFO org.apache.pig.tools.pigstats.ScriptState - Pig features used in the script
: GROUP_BY, ORDER_BY, FILTER
2026-09-08 17:57:46,292 [main] INFO org.apache.pig.newplan.logical.optimizer.LogicalPlanOptimizer - {RULES_ENAB
LED=[AddForEach, ColumnMapKeyPrune, ConstantCalculator, GroupByConstParallelSetter, LimitOptimizer, LoadTypeCast
Insert, MergeFilter, MergeForEach, NestedLimitOptimizer, PartitionFilterOptimizer, PredicatePushdownOptimizer,
PushDownForEachFlatten, PushUpFilter, SplitFilter, StreamTypeCastInsert]}
2026-09-08 17:57:46,330 [main] INFO org.apache.pig.impl.util.SpillableMemoryManager - Selected heap (PS Old Gen
) of size 699400192 to monitor. collectionUsageThreshold = 489580128, usageThreshold = 489580128
2026-09-08 17:57:46,378 [main] INFO org.apache.pig.newplan.logical.rules.ColumnPruneVisitor - Columns pruned fo
r retail: $1, $2, $6, $7
2026-09-08 17:57:46,461 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MRCompiler - F
ile concatenation threshold: 100 optimistic? false
2026-09-08 17:57:46,554 [main] INFO org.apache.pig.backend.hadoop.executionengine.util.CombinerOptimizerUtil -
Choosing to move algebraic foreach to combiner
2026-09-08 17:57:46,575 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.SecondaryKeyOp
timizerMR - Using Secondary Key Optimization for MapReduce node scope-78
2026-09-08 17:57:46,586 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOpti
mizer - MR plan size before optimization: 3
2026-09-08 17:57:46,586 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOpti
```

```
YTRUN-10_pig_monthly_sales_trend

rcCalculatorProcessTree : org.apache.hadoop.yarn.util.WindowsBasedProcessTree@5d8bcc0c
2026-09-08 17:57:56,212 [LocalJobRunner Map Task Executor #0] INFO org.apache.hadoop.mapred.MapTask - Processin
g split: Number of splits :1
Total Length = 397
Input split[0]:
Length = 397
ClassName: org.apache.hadoop.mapreduce.lib.input.FileSplit
Locations:
-----
2026-09-08 17:57:56,214 [LocalJobRunner Map Task Executor #0] INFO org.apache.pig.backend.hadoop.executionengin
e.mapReduceLayer.PigRecordReader - Current split being processed file:/D:/Desktop/youtube_hadoop_project/pig/tmp
/temp1294126528/tmp-1130651298/part-r-00000:0+397
2026-09-08 17:57:56,222 [LocalJobRunner Map Task Executor #0] INFO org.apache.hadoop.mapred.MapTask - (EQUATOR)
0 kv1 26214396(104857584)
2026-09-08 17:57:56,223 [LocalJobRunner Map Task Executor #0] INFO org.apache.hadoop.mapred.MapTask - mapreduce
.task.io.sort.mb: 100
2026-09-08 17:57:56,223 [LocalJobRunner Map Task Executor #0] INFO org.apache.hadoop.mapred.MapTask - soft limi
t at 83886080
2026-09-08 17:57:56,223 [LocalJobRunner Map Task Executor #0] INFO org.apache.hadoop.mapred.MapTask - bufstart
= 0; bufvoid = 104857600
2026-09-08 17:57:56,223 [LocalJobRunner Map Task Executor #0] INFO org.apache.hadoop.mapred.MapTask - kvstart =
26214396; length = 6553600
2026-09-08 17:57:56,226 [LocalJobRunner Map Task Executor #0] INFO org.apache.hadoop.mapred.MapTask - Map output
collector class = org.apache.hadoop.mapred.MapTask$MapOutputBuffer
2026-09-08 17:57:56,230 [LocalJobRunner Map Task Executor #0] INFO org.apache.pig.impl.util.SpillableMemoryMana
ger - Selected heap (PS Old Gen) of size 699400192 to monitor. collectionUsageThreshold = 489580128, usageThresh
old = 489580128
2026-09-08 17:57:56,231 [LocalJobRunner Map Task Executor #0] WARN org.apache.pig.data.SchemaTupleBackend - Sch
emaTupleBackend has already been initialized
2026-09-08 17:57:56,232 [LocalJobRunner Map Task Executor #0] INFO org.apache.pig.backend.hadoop.executionengin
e.mapReduceLayer.PigGenericMapReduce$Map - Aliases being processed per job phase (AliasName[line,offset]): M: mo
nthly_sorted[20,17] C: R:
2026-09-08 17:57:56,234 [LocalJobRunner Map Task Executor #0] INFO org.apache.hadoop.mapred.LocalJobRunner -
2026-09-08 17:57:56,235 [LocalJobRunner Map Task Executor #0] INFO org.apache.hadoop.mapred.MapTask - Starting
flush of map output
2026-09-08 17:57:56,235 [LocalJobRunner Map Task Executor #0] INFO org.apache.hadoop.mapred.MapTask - Spilling
map output
2026-09-08 17:57:56,235 [LocalJobRunner Map Task Executor #0] INFO org.apache.hadoop.mapred.MapTask - bufstart
= 0; bufend = 384; bufvoid = 104857600
2026-09-08 17:57:56,235 [LocalJobRunner Map Task Executor #0] INFO org.apache.hadoop.mapred.MapTask - kvstart =
26214396(104857584); kvend = 26214348(104857392); length = 49/6553600
```

YTRUN-10\_pig\_monthly\_sales\_trend

2026-09-08 17:57:56,845 [main] INFO org.apache.pig.Main - Pig script completed in 13 seconds and 111 milliseconds (13111 ms)

D:\Desktop\bdal\_project>type D:\Desktop\bdal\_project\PigAnalysis\output\MonthlySalesTrend\part-r-00000

|         |            |       |      |
|---------|------------|-------|------|
| 2010-12 | 823746.14  | 41480 | 1559 |
| 2011-01 | 691364.56  | 34306 | 1086 |
| 2011-02 | 523631.89  | 27105 | 1100 |
| 2011-03 | 717639.36  | 35803 | 1454 |
| 2011-04 | 537808.62  | 29096 | 1246 |
| 2011-05 | 770536.02  | 36164 | 1681 |
| 2011-06 | 761739.9   | 35977 | 1533 |
| 2011-07 | 719221.19  | 38645 | 1475 |
| 2011-08 | 759138.38  | 34483 | 1361 |
| 2011-09 | 1058590.17 | 49261 | 1837 |
| 2011-10 | 1154979.3  | 59304 | 2040 |
| 2011-11 | 1509496.33 | 83369 | 2769 |
| 2011-12 | 638792.68  | 25111 | 819  |